

Question Paper

Exam Date & Time: 28-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Software Engineering [ECM_3124]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Considering an online ticket booking system, explain how requirement analysis helps address challenges like concurrent booking, payment processing, and seat selection. How would you gather user and stakeholder requirements to ensure the system is reliable, user-friendly, and scalable? (5)
- A) Course Outcome: "CO2" / Cognitive Level: "Understand"
- B) What is Requirement Gathering and Analysis in software engineering, and why is it critical to project success? (5)
- Course Outcome: "CO2" / Cognitive Level: "Remember"
- 2) Considering a banking system, describe how design process steps guide you from requirements to architecture and module design. How would you ensure the design exhibits strong cohesion within modules and loose coupling across them to promote security, reliability, and ease of testing? What design approach is most suitable for handling complex transaction processing and compliance rules in this system? (5)
- A) Course Outcome: "CO3" / Cognitive Level: "Analysis"
- B) What is meant by layered arrangement of modules, and how does it benefit software development? (5)
- Course Outcome: "CO3" / Cognitive Level: "Remember"
- 3) Describe what a Data Flow Diagram (DFD) is and its role in system analysis (5)
- A) Course Outcome: "CO4" / Cognitive Level: "Remember"
- B) For a banking system, illustrate how detailed design builds on the structured analysis and design phases by specifying module interfaces, data structures, and algorithm details. How would you document this detailed design, and what role does design review play in ensuring system reliability and compliance? (5)
- Course Outcome: "CO4" / Cognitive Level: "Analysis"
- 4) In the context of a banking system, explain how Object-Oriented Analysis and Design (OOAD) methodology aids in building a robust software model. How can design patterns improve maintainability and scalability in this case? (5)
- A) Course Outcome: "CO4" / Cognitive Level: "Understand"

B) What is a Class Diagram in UML, and how does it represent the static structure of a system? (5)

Course Outcome: "CO4" / Cognitive Level: "Remember"

5) In the coding phase for a library management system, how would you implement unit testing to verify individual modules such as book catalog and user authentication? Explain the importance of code reviews and software documentation in ensuring code quality and maintainability. (5)

A)

Course Outcome: "CO5" / Cognitive Level: "Application"

B) What is Unit Testing, and why is it important in the software development lifecycle? (5)

Course Outcome: "CO5" / Cognitive Level: "Remember"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 ECM_3124:Software Engineering

#	Course Outcome	Description	Marks
1	2	CO 2	10
2	3	CO 3	10
3	4	CO 4	20
4	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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#	Level	No of Questions	Marks (%)
1	Remember	5	25(50%)
2	Understand	2	10(20%)
3	Application	1	5(10%)
4	Analysis	2	10(20%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)